ETH zürich GHE GLOBAL HEALTH ENGINEERING

Mzuzu University
Service Perfection Self-Reliance

MALAWI UNIVERSITY of BUSINESS AND APPLIED SCIENCES

Energy and Device Data Collection in Healthcare Facilities in Malawi

Gian Teufen & Till Häussner

Research proposal presentation

Rumphi District Health Office, Malawi

D MAVT

Energy and Device Data Collection in Healthcare Facilities in Malawi

Principal Investigators (PI)

Name	Group/Chair/Institute	University
Mr. Gian Teufen	Global Health Engineering	ETH Zurich
Mr. Till Häussner	Global Health Engineering	ETH Zurich

Involved Researchers

Name	Group/Chair/Institute	University
Dr. Frank Mnthambala	Agrisciences Department	Mzuzu University
Mr. Hope Kelvin Chilunga	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Dr. Rebecca Alcock	Global Health Engineering, Center for Global Engineering	ETH Zürich, University of Toronto
Prof. Dr. Elizabeth Tilley	Global Health Engineering	ETH Zurich
Dr. Jakub Tkaczuk	Global Health Engineering	ETH Zürich

Electrification of Healthcare Facilities

- **Almost 1 billion people worldwide served by HCFs without reliable electricity access.**
  - **at 641 million, vast majority in Sub-Saharan Africa.**

(World Health Organization et al., 2023)

- **Health service provision suffers greatly without solid electrification.**

(Shastry & Rai, 2021; WHO, 2015)

- **Electrification programs can have a significant and lasting positive impact.**

(Chen et al., 2019; Ramji et al., 2017; Irwin et al., 2020; Khogali et al., 2022)

Electrification of Healthcare Facilities in Malawi

- **82% of Malawians live in rural areas, making electrification very difficult.**

(Ehimen et al., 2023; Rahman et al., 2024; World Bank Open Data, n.d.)

- **Rural, primary-care HCFs least likely to have stable grid access, many relying on other sources.**

(Malawi HHFA 2018/2019, 2019; Reuland et al., 2020; Ngongoma et al., 2025)

- **Very little data on electricity supply and needs, medical devices and amenities.**

(Suhlrie et al., 2018; World Health Organization et al., 2023; Reuland et al., 2020)

Decentralized Solar Microgrids

- **Renewable, off-grid energy systems play a crucial role in rapid, low-cost electrification with great climate resilience.**

(Kassam, 2024; World Health Organization et al., 2023; Ramji et al., 2017)

- **Data needed on systems size, demands and locations for effective system design and implementation.**

(Moner-Girona et al., 2021; Rahman et al., 2024; World Health Organization et al., 2023)

Operations & Maintenance of Energy Systems in Malawi

- **Energy systems in HCFs in Sub-Saharan Africa often lack sufficient capacity or are underfunded for operations & maintenance.**

  (UNICEF, 2018)

- **74% of all health financing in Malawi comes from donors, making long-term support volatile.**

  (Borghi et al., 2018)

- **Proper O&M of energy systems crucial for long-term success.**

  (World Health Organization et al., 2023)

Objectives of this Study

Collect detailed data on:

- Power supply
- Electrical appliances and medical devices
- Funding and procurement
- O&M of energy system
- Facility service readiness

… in small Malawian healthcare facilities …

… to inform:

- Policy makers & officials
- Facility managers
- Energy planners and engineers
- NGOs
- and others

… with a systematic and detailed countrywide overview.

Type of Healthcare Facility

Focus:

rural, government-run health centres

Facility Sampling

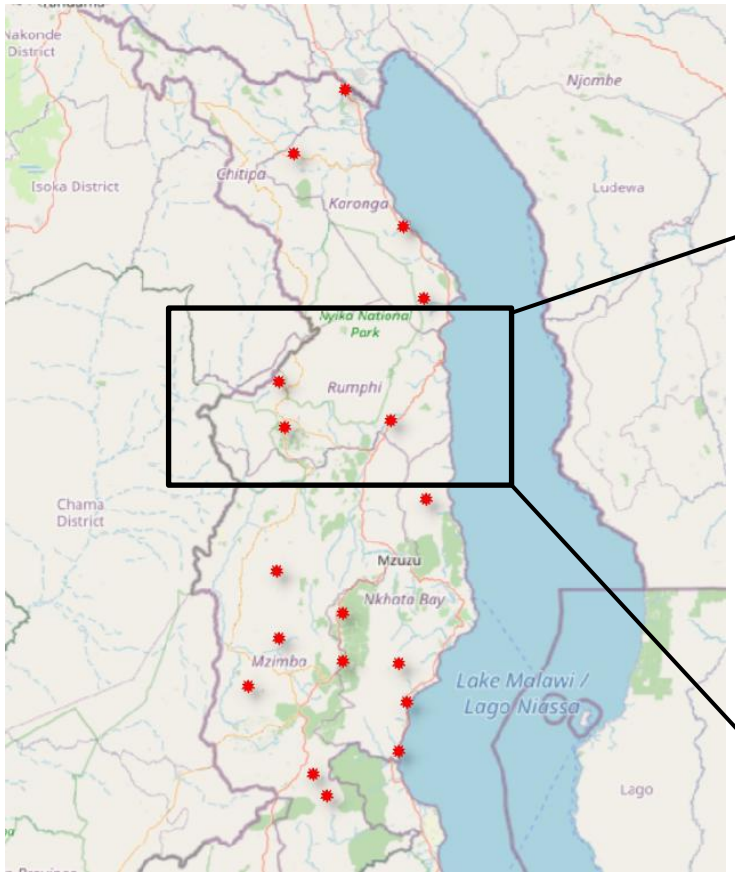
Purposive Sampling

- Facilities with diverse information
- Wide geographic coverage

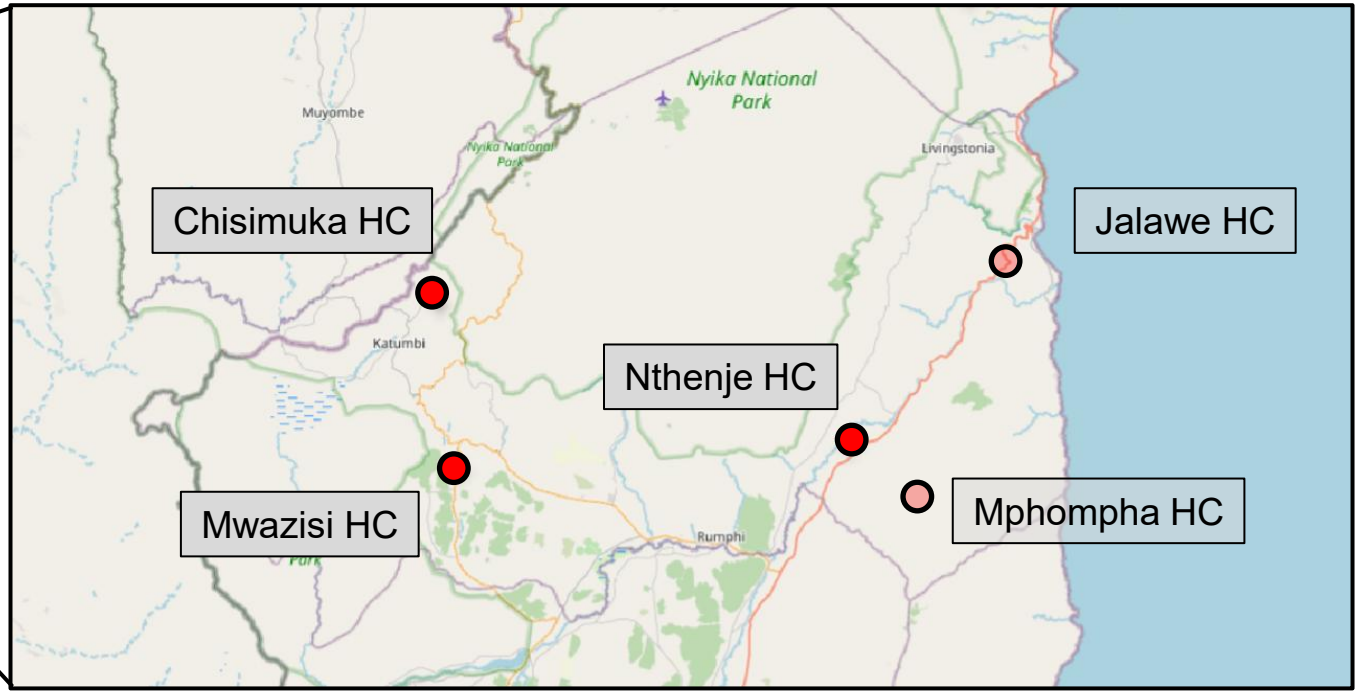
Inclusion Criteria

- Health centres
- Government-run
- Electrified
- Grid and off-grid
- 3-6 facilities per district in the north

Facility Sampling



Rumphi Health Centres



Methods for Data Collection

A) Electrical Device Inspection

- Medical Devices
- Electrical Appliances

Data collected per device:

- Device name
- Device type
- Energy Characteristics
- Condition
- Procurement
- Maintenance

B) Structured Questionnaire

Sections:

0. Facility Identifiers
1. Energy System
2. Origins, Operations and Maintenance
3. Climate Change & Weather Impact
4. Amenities & Equipment

Tools for Data Collection

A) Electrical Device Inspection

- **Microsoft Excel / Google Sheets**
- **SEM5000E Energy Logger**

B) Structured Questionnaire

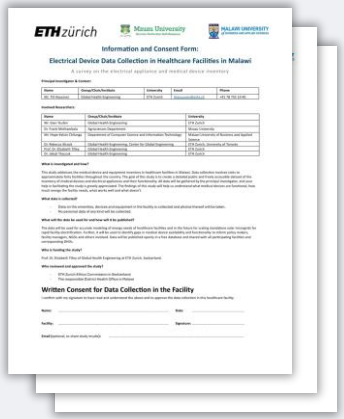
- **Kobo Toolbox**
  - In English, Chichewa and Chitumbuka
- **Microsoft Excel / Google Sheets**

KoboToolbox

Information and Consent Forms for Data Collection

A) Electrical Device Inspection

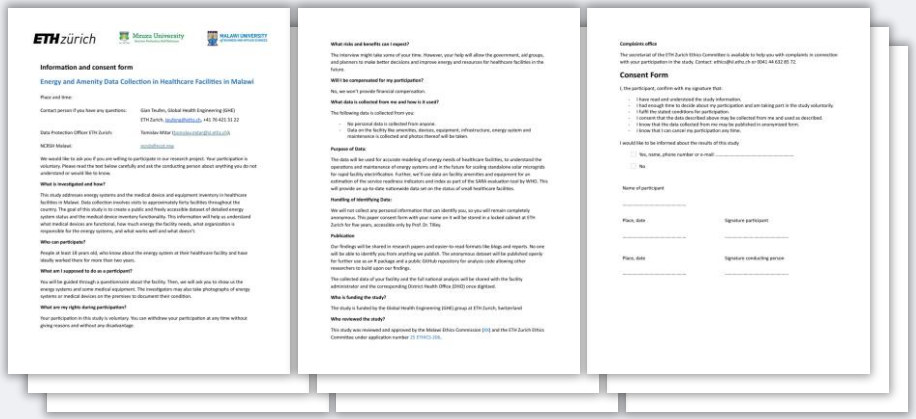
- for Facility Representative



In English, Chichewa & Chitumbuka

B) Structured Questionnaire

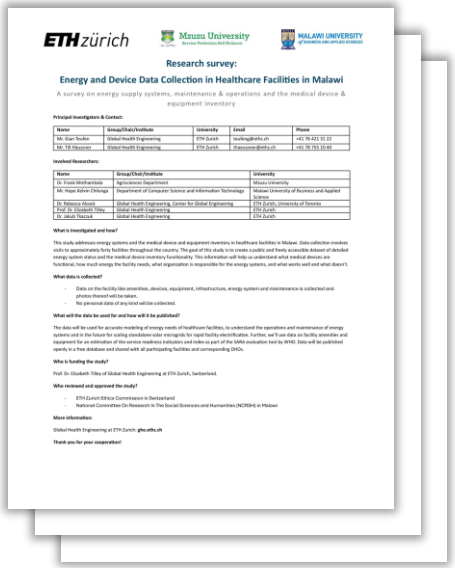
- for Participant



In English, Chichewa & Chitumbuka

Study Information Sheet

Study Information Sheet
for entire facility and all interested persons



Research survey:

Energy and Device Data Collection in Healthcare Facilities in Malawi

A survey on energy supply systems, maintenance & operations and the medical device & equipment inventory

Principal Investigators & Contact:

Name	Group/Chair/Institute	University	Email	Phone
Mr. Gian Teufen	Global Health Engineering	ETH Zurich	teufeng@ethz.ch	+41 76 421 31 22
Mr. Till Häusner	Global Health Engineering	ETH Zurich	thaeusner@ethz.ch	+41 78 753 10 40

Involved Researchers:

Name	Group/Chair/Institute	University
Dr. Frank Mkhwanbola	Agriculturecs Department	Mzuzu University
Mr. Hope Kelvin Chilunga	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Dr. Rebecca Alcock	Global Health Engineering, Center for Global Engineering	ETH Zurich, University of Toronto
Prof. Dr. Elizabeth Tilley	Global Health Engineering	ETH Zurich
Dr. Jakob Tkaczuk	Global Health Engineering	ETH Zurich

What is investigated and how?

This study addresses energy systems and the medical device and equipment inventory in healthcare facilities in Malawi. Data collection involves visits to approximately forty facilities throughout the country. The goal of this study is to create a public and freely accessible dataset of detailed energy system status and the medical device inventory functionality. This information will help us understand what medical devices are functional, how much energy the facility needs, what organization is responsible for the energy systems, and what works well and what doesn't.

What data is collected?

- Data on the facility like amenities, devices, equipment, infrastructure, energy system and maintenance is collected and photos thereof will be taken.
- No personal data of any kind will be collected.

What will the data be used for and how will it be published?

The data will be used for accurate modeling of energy needs of healthcare facilities, to understand the operations and maintenance of energy systems and in the future for scaling standalone solar microgrids for rapid facility electrification. Further, we'll use data on facility amenities and equipment for an estimation of the service readiness indicators and index as part of the SARA evaluation tool by WHO. Data will be published openly in a free database and shared with all participating facilities and corresponding DHOs.

Who is funding the study?

Prof. Dr. Elizabeth Tilley of Global Health Engineering at ETH Zurich, Switzerland.

Who reviewed and approved the study?

- ETH Zurich Ethics Commission in Switzerland
- National Committee On Research In The Social Sciences and Humanities (NCRSH) in Malawi

More information:

Global Health Engineering at ETH Zurich: **ghe.ethz.ch**

Thank you for your cooperation!

In English, Chichewa & Chitumbuka

Timeline

2025

August

North

Mzimba
Nkhata Bay
Rumphi
Karonga

September

South

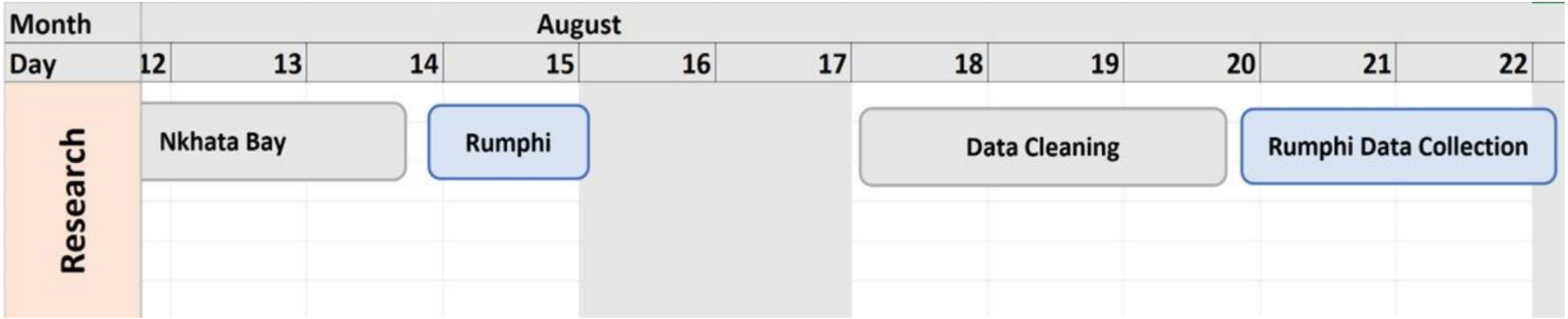
Machinga
Mangochi
Mulanje
Mwanza
Neno
Nsanje
…

October

Central

Dowa
Lilongwe
Mchinji
Ntchisi
Kasungu
Dedza
…

Timeline


Month
August
Day
12
13
14
15
16
17
18
19
20
21
22
Research
Nkhata Bay
Rumphi
Data Cleaning
Rumphi Data Collection

Important Ethical Aspects: Informed Consent and Debriefing

- **Recruitment of Participants** > DHOs/HCFs
- **Dissemination of Study Information** > Info Sheets, Informed Consent Forms
- **Expression of Informed Consent** > Signature on Informed Consent Forms
- **Storage of Consent Forms** > Locked at ETH Zurich for 5 years
- **Sharing of Study Results** > Report to DHOs/HCFs

Important Ethical Aspects: Data Handling

- **Data Collection** > NO personal data collected in any form
- **Data Storage** > Locked Cabinet, ETH Servers
- **Data Analysis** > R/RStudio, Python, QGIS, Excel, Google Sheets
- **Data Management** > Git/GitHub, PHP, SQL Databases
- **Publication** > Freely and accessibly,
  according to open science principles

Important Ethical Aspects: Risks, Cancellations, Consequences

- **Risks** > NO personal physical or psychological risks
- **Discomforts** > Time burden
- **Interruptions** > Interruptions to facility proceedings avoided
- **Cancellation** > any time, without reasons or consequences

Study Results

- **Detailed, comprehensive, nationwide HCF data on:**
  - Electricity access, supply and demands
  - Medical devices and electrical appliances
  - Funding and procurement of appliance
  - SARA service readiness indices
- **Published openly for future work.**

To conclude

Let's help Malawian healthcare succeed!

Our deepest gratitude to all for listening. ☺

ETH zürich

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Appendix 1:

Letter Head by Dr. Frank Mnthambala

OFFICE OF THE HEAD OF DEPARTMENT

AGRISCIENCES

Mzuzu University
Private Bag 201
Luwinga
Mzuzu 2
Malawi
Tel.: (265) 01 320 722/575
Fax: (265) 01 320 568
E-mail: agrisciences@mzuni.ac.mw

6/6/2025

Dear DHO,
I write to request that you provide access to your health facilities to Gian Teufen and Till Haeussner who are working with me on a health clinic electrification project.

Specifically, Mr. Teufen will be focused on developing an energy systems, operations and maintenance dataset of Malawi's health posts in which facility-level data on the energy supply status will be collected. This means documenting the energy source mix and energy storage by writing down nameplate characteristics and inspecting functionality and installation. Additionally, the origins and current operations and maintenance of energy systems and key medical devices will be documented by talking to the responsible persons. **The goal is then for this dataset to enable accountability among associated donors or implementing partners for implemented energy systems**.

Mr. Haeussner will be developing an open database of electrical medical devices for health posts in Malawi. Appropriately sizing a renewable energy system for healthcare facilities requires estimates on the energy load of the electrical devices present. The goal of this work is to start an open database of the electrical devices being used in Malawi's health posts. For this, data on the electrical device inventory will be collected, including brand, name, energy characteristics and functionality. Additionally, the power consumption profile of certain devices will be measured using a conventional energy logger. **The results of this research will guide more accurate load estimations in healthcare facility electrification efforts throughout Malawi and beyond.**

Thank you for your consideration,

Dr. Frank Mnthambala

Head of Agrisciences Department.

Tel.: (265) 01 320 722/575
Fax: (265) 01 320 568

agrisciences@mzuni.ac.mw

Letter to HCFs

From:

Mzimba District Health Office

To whom it might concern,

This is to confirm that the following people will be doing approved research in healthcare facilities in this district.

Principal Investigators (PI):

Name	Group/Chair/Institute	University	Email	Phone
Mr. Gian Teufen	Global Health Engineering	ETH Zurich	teufeng@ethz.ch	+41 76 421 31 22
Mr. Till Häussner	Global Health Engineering	ETH Zurich	thaeussner@ethz.ch	+41 78 755 10 40

Involved Researchers:

Name	Group/Chair/Institute	University
Dr. Frank Mnthambala	Agrisciences Department	Mzuzu University
Mr. Hope Kelvin Chilunga	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Dr. Rebecca Alcock	Global Health Engineering, Center for Global Engineering	ETH Zurich, University of Toronto
Prof. Dr. Elizabeth Tilley	Global Health Engineering	ETH Zurich
Dr. Jakub Tkaczuk	Global Health Engineering	ETH Zurich

The goal of this research is to collect data on the energy supply and needs of health posts to inform accurate sizing of new renewable energy systems for better healthcare facility electrification.

The District Health Officer respectfully requests your cooperation in:

- Facilitating access to all relevant equipment, rooms and installations.
- Introducing the researchers to personnel that can provide accurate information on the systems in use.

The researchers will collect the following data in facility:

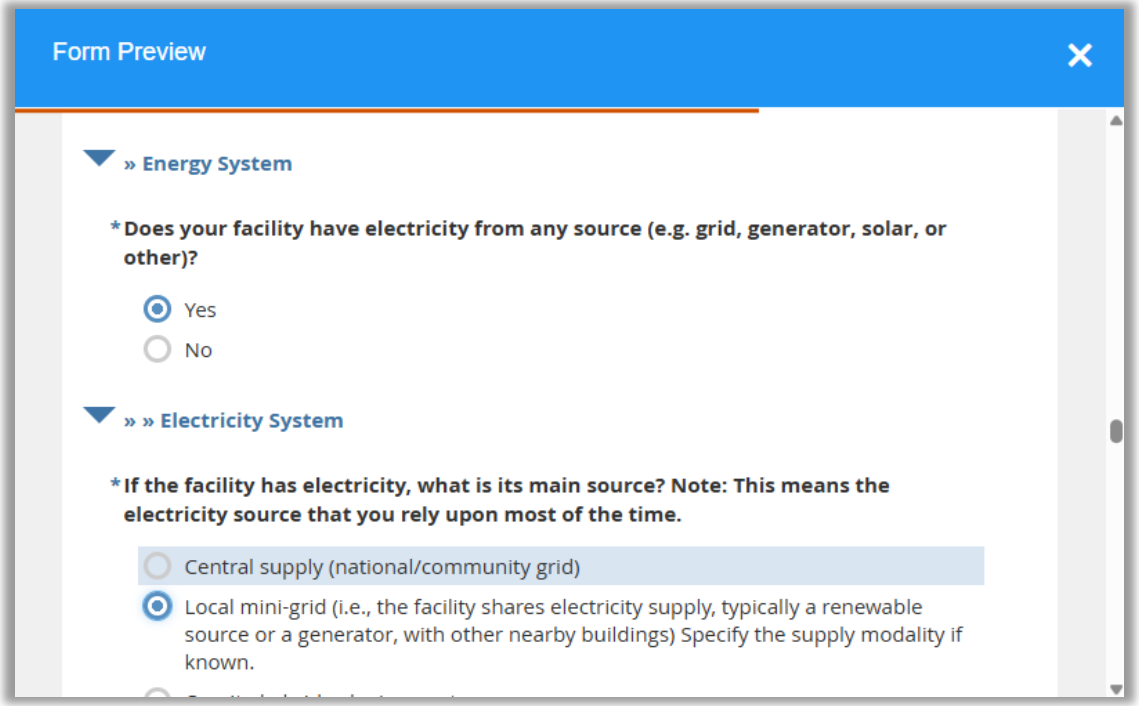
- Energy supply: energy mix, functionality, nameplate characteristics, installation, operations and maintenance. This involves a visual inspection as well as an interview with responsible personnel.
- Electrical devices: brand, name, nameplate energy characteristics, functionality and quantity. For some devices, power consumption will be measured using a conventional energy logger.
- Amenities and equipment: detailed availability data as directed by the SARA evaluation tool by WHO.

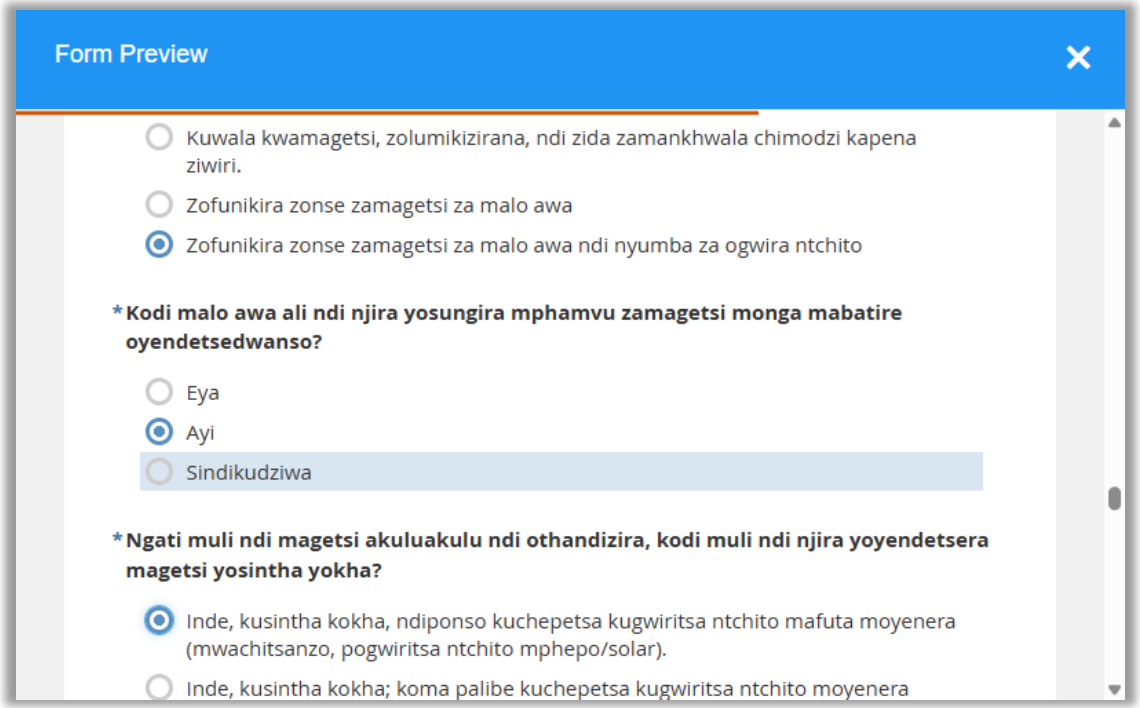
The researchers will NOT collect any personal data of patients or staff. Further, they will not interfere with ongoing consultations and the workings of the facility in general.

We are thankful for your help and collaboration with this project.

DATE: **NAME**: **SIGNATURE:**

Appendix 3: Kobo Toolbox Screenshots

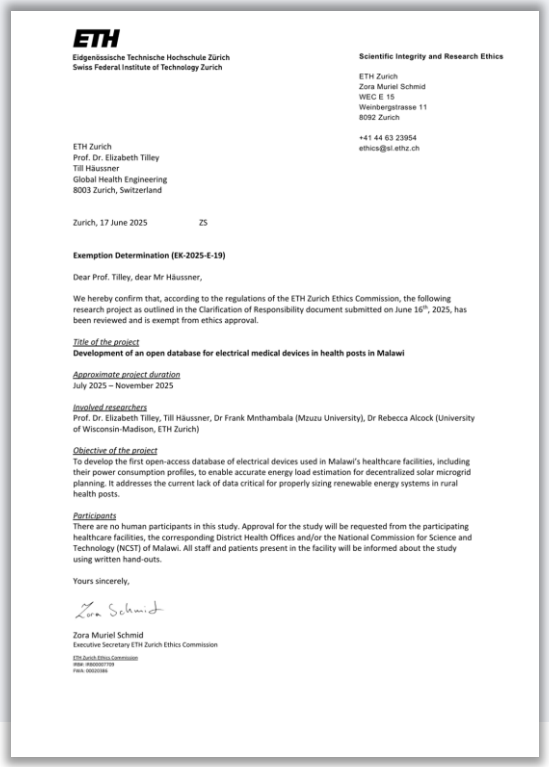




Appendix 4: ETH Zurich Ethics Committee Approvals

A) Electrical Device Inspection

Waiver *EK-2025-E-19*



ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Scientific Integrity and Research Ethics

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+41 44 63 23954
ethics@sl.ethz.ch

ETH Zurich
Prof. Dr. Elizabeth Tilley
Till Häussner
Global Health Engineering
8003 Zurich, Switzerland

Zurich, 17 June 2025 ZS

Exemption Determination (EK-2025-E-19)

Dear Prof. Tilley, dear Mr Häussner,

We hereby confirm that, according to the regulations of the ETH Zurich Ethics Commission, the following research project as outlined in the Clarification of Responsibility document submitted on June 16th, 2025, has been reviewed and is exempt from ethics approval.

Title of the project
Development of an open database for electrical medical devices in health posts in Malawi

Approximate project duration
July 2025 – November 2025

Involved researchers
Prof. Dr. Elizabeth Tilley, Till Häussner, Dr Frank Mnthambala (Mzuzu University), Dr Rebecca Alcock (University of Wisconsin-Madison, ETH Zurich)

Objective of the project
To develop the first open-access database of electrical devices used in Malawi's healthcare facilities, including their power consumption profiles, to enable accurate energy load estimation for decentralized solar microgrid planning. It addresses the current lack of data critical for properly sizing renewable energy systems in rural health posts.

Participants
There are no human participants in this study. Approval for the study will be requested from the participating healthcare facilities, the corresponding District Health Offices and/or the National Commission for Science and Technology (NCST) of Malawi. All staff and patients present in the facility will be informed about the study using written hand-outs.

Yours sincerely,

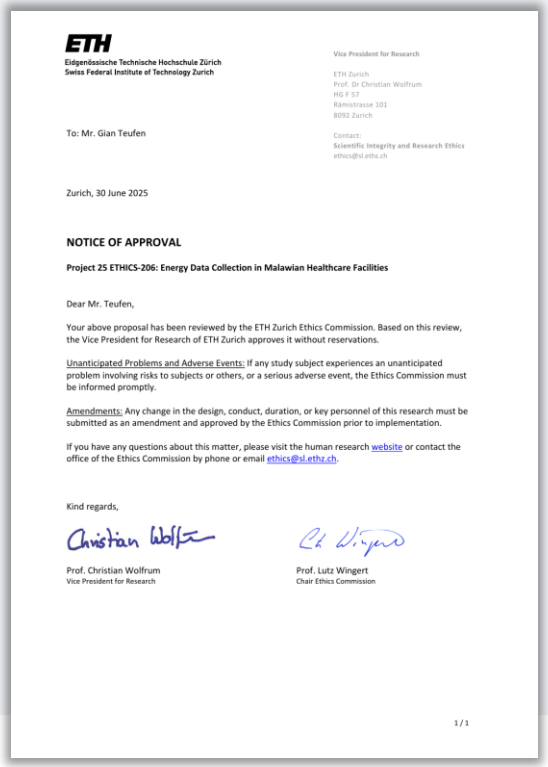
Zora Schmid

Zora Muriel Schmid
Executive Secretary ETH Zurich Ethics Commission

ETH Zurich Ethics Commission
PNR: PN00007709
FKN: 00020586

B) Structured Questionnaire

Approval *25 ETHICS-206*



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Swiss Federal Institute of Technology Zurich

Vice President for Research

ETH Zurich
Prof. Dr Christian Wolfrum
HG F 57
Rämistrasse 101
8092 Zurich

Contact:
Scientific Integrity and Research Ethics
ethics@sl.ethz.ch

To: Mr. Gian Teufen

Zurich, 30 June 2025

NOTICE OF APPROVAL

Project 25 ETHICS-206: Energy Data Collection in Malawian Healthcare Facilities

Dear Mr. Teufen,

Your above proposal has been reviewed by the ETH Zurich Ethics Commission. Based on this review, the Vice President for Research of ETH Zurich approves it without reservations.

Unanticipated Problems and Adverse Events: If any study subject experiences an unanticipated problem involving risks to subjects or others, or a serious adverse event, the Ethics Commission must be informed promptly.

Amendments: Any change in the design, conduct, duration, or key personnel of this research must be submitted as an amendment and approved by the Ethics Commission prior to implementation.

If you have any questions about this matter, please visit the human research website or contact the office of the Ethics Commission by phone or email ethics@sl.ethz.ch.

Kind regards,

Christian Wolfrum

Prof. Christian Wolfrum
Vice President for Research

Prof. Lutz Wingert
Chair Ethics Commission

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